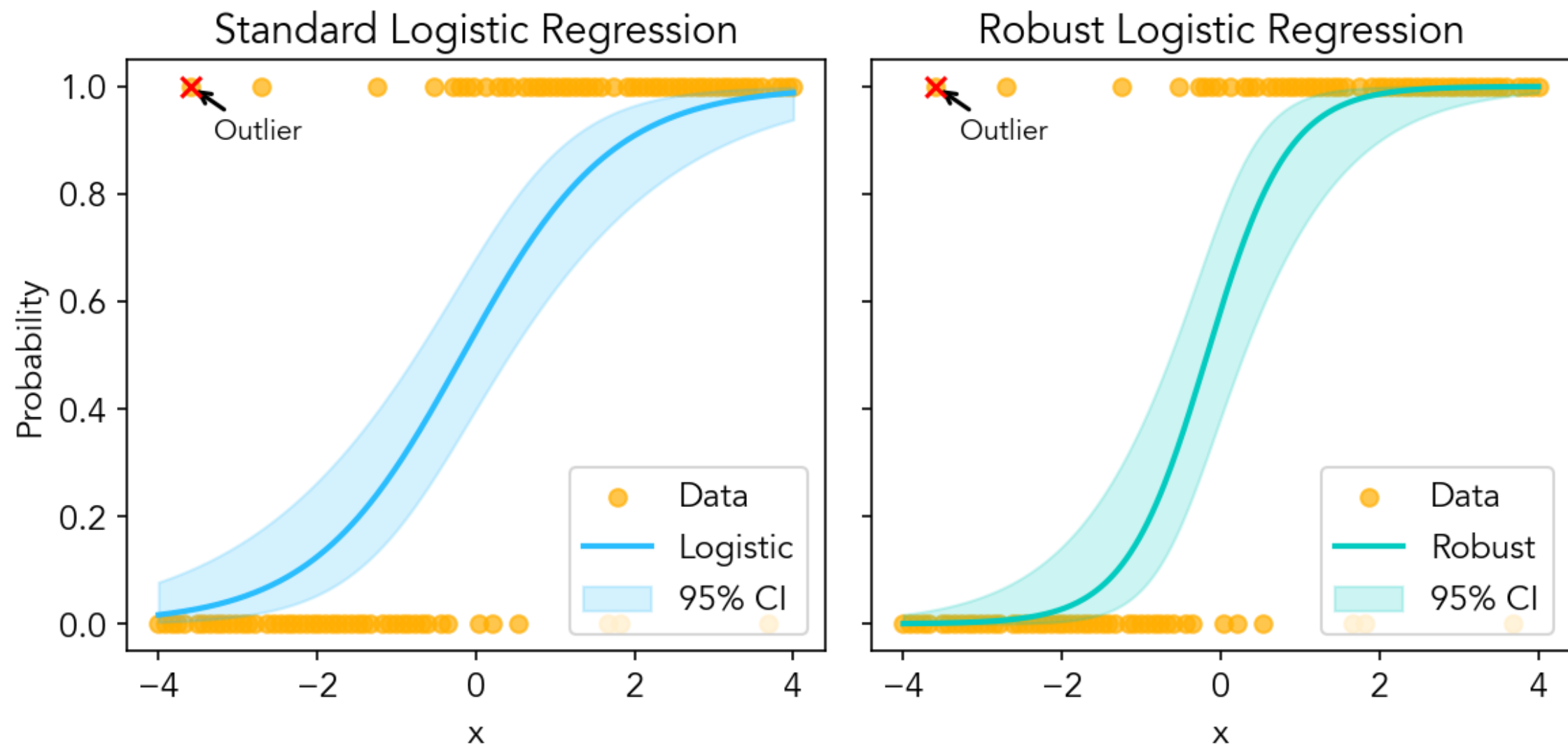


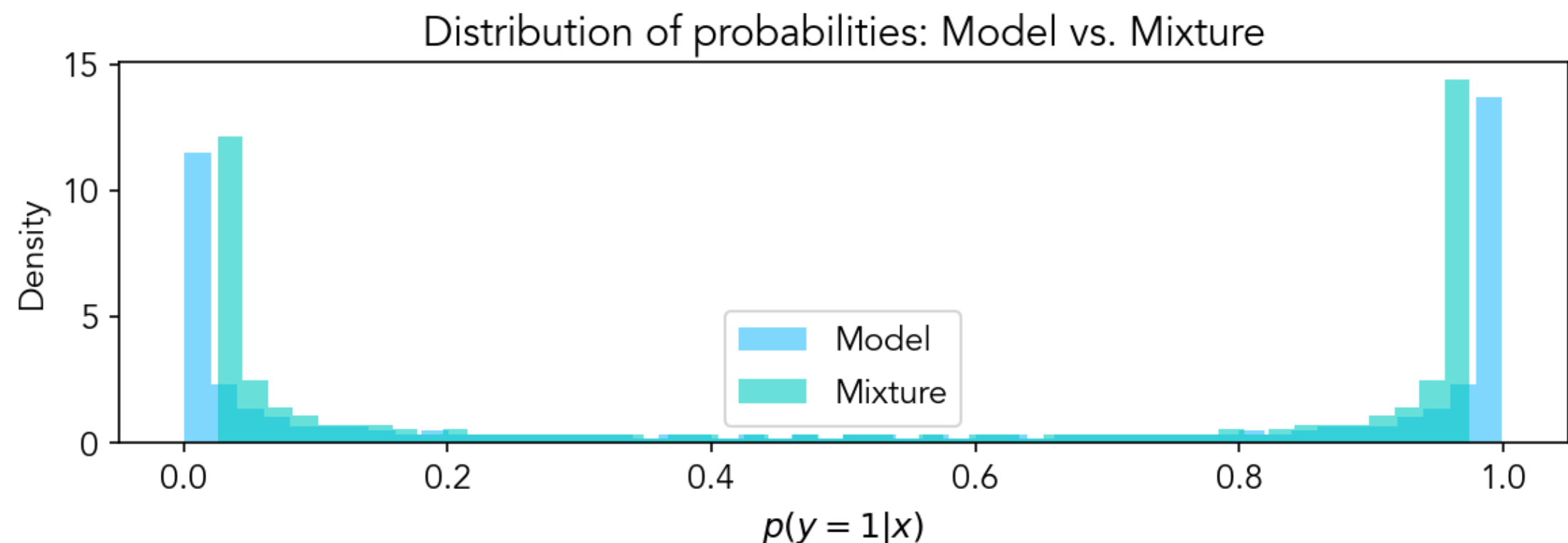
Robust Logistic Regression



Outliers in labels (label noise) can skew the results of logistic regression modeling. One solution to reduce the impact of contamination on estimates is to use a robust version of logistic regression¹.



This model works by introducing the mixture of the two “sources” of labels, the model and the random (outlier) process. The robust probability curve is pulled toward 0.5 where the model is uncertain, making the fit more robust to the presence of outliers.



1. Kevin P. Murphy. Probabilistic Machine Learning: An Introduction, MIT, 2022.

Robust Logistic Regression



Below we provide a sketch implementation of mixture log likelihood with Fisher scoring with Python 🐍¹:

```
import numpy as np
from scipy.special import expit as sigmoid

def robust_logistic_regression(
    X, y, epsilon=0.05, max_iter=30, tol=1e-6
):
    """Robust Logistic Regression."""
    n_features = X.shape[1]
    beta = np.zeros(n_features)
    q = 0.5 * np.ones_like(y, dtype=float)
    for _ in range(max_iter):
        z = X @ beta      # Logit
        p = sigmoid(z)    # Inverse logit

        # Assign probabilities to true label
        p_model = np.where(y == 1, p, 1 - p)

        # Calculate weights
        numerator = (1 - epsilon) * p_model
        denominator = numerator + epsilon * q
        w = numerator / denominator

        # Calculate weights and score
        W_diag = w * p * (1 - p)
        score = X.T @ (w * (y - p))

        # Calculate weighted Fisher information
        fisher_info = X.T @ (W_diag[:, np.newaxis] * X)

        try: # Update step
            beta_new = np.linalg.inv(fisher_info) @ score
        except np.linalg.LinAlgError:
            beta_new = np.linalg.pinv(fisher_info) @ score
        beta += beta_new
        if np.linalg.norm(beta_new) < tol: # Convergence
            break
    return beta
```

1. Here we provide the Fisher Scoring algorithm example, which leverages Fisher information to estimate the coefficients, beta. Other alternatives to fitting are SGD (using only first-order gradients) and Bayesian logistic regression with pymc. Notebook with examples: <https://lnkd.in/esHjSmZZ>

Robust Logistic Regression



The fisher-scoring package provides fast maximum likelihood Python 🐍 implementations with standard errors, confidence intervals, and rich summaries for:

Proportions

- Logistic Regression
- Multinomial Logistic Regression

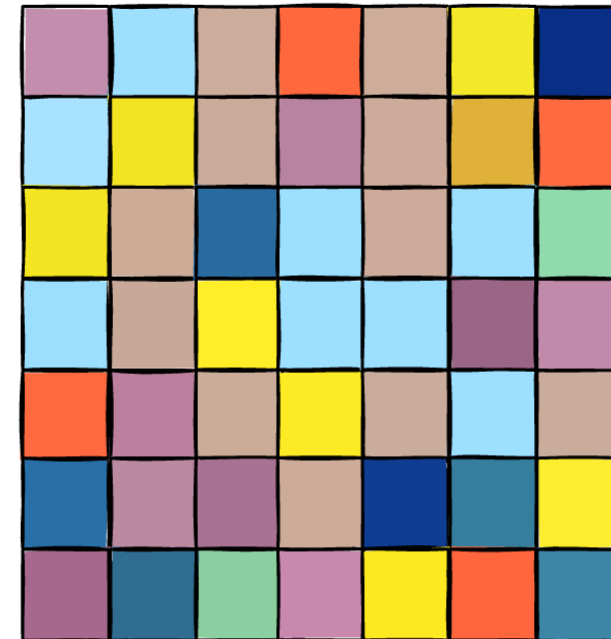
Proportions (imbalance and outliers)

- Focal Loss Logistic Regression
- Robust Logistic Regression

Count data

- Poisson Regression
- Negative Binomial Regression

FISHER SCORING
WITH PYTHON



Try robust logistic regression with fisher-scoring package ↘

```
%pip install fisher-scoring==2.0.5
from fisher_scoring import RobustLogisticRegression

rlr = RobustLogisticRegression()
rlr.fit(X, y)

rlr.display_summary()
```

<https://github.com/xRiskLab/fisher-scoring>